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Stability preserving model reduction for linearly coupled linear time-invariant systems

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We develop a stability preserving model reduction method for linearly coupled linear time-invariant (LTI) systems. The method extends the work of Monshizadeh et al. for multi-agent systems with identical LTI agents. They propose using Bounded Real Balanced Truncation to preserve a sufficient condition for stability of the coupled system. Here, we extend this idea to arbitrary linearly coupled LTI systems using the sufficient condition for stability introduced by Reis and Stykel. The model reduction error bounds for this method also follow from results of Reis and Stykel, which allows the adaptive choice of reduced orders. We demonstrate the method on Reis's and Stykel's coupled string-beam example.

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1 Introduction

In this paper, we consider the reduction of coupled systems, in particular of linear time-invariant (LTI) systems linearly coupled through their inputs and outputs. Moreover, we want to preserve the interconnection structure by only reducing the subsystems, while also preserving the stability of the coupled system. We extend the stability preserving model reduction method for multi-agent systems from [1] to coupled systems considered in [2].

2 The model reduction method

Reis and Stykel [2] study systems of k coupled LTI subsystems

$$E_i\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t), \quad y_i(t) = C_i x_i(t), \tag{1a}$$

with coupling, external input, and external output

$$u_i(t) = K_{i1}y_1(t) + \cdots + K_{ik}y_k(t) + H_i u(t), \quad y(t) = R_1 y_1(t) + \cdots + R_k y_k(t), \tag{1b}$$

where $i \in \{1, 2, \ldots, k\}$, $x_i(t) \in \mathbb{R}^{n_i}$, $u_i(t) \in \mathbb{R}^{m_i}$, $y_i(t) \in \mathbb{R}^{p_i}$, $u(t) \in \mathbb{R}^m$, and $y(t) \in \mathbb{R}^p$. Here, we assume the matrices E_i to be nonsingular and that the matrix pencils $A_i - \lambda E_i$ are stable. The LTI system (1) can be written [2] as

$$\mathcal{E}\dot{x}(t) = \mathcal{A}x(t) + \mathcal{B}u(t), \quad y(t) = \mathcal{C}x(t), \tag{2}$$

with $\mathcal{E}, \mathcal{A} \in \mathbb{R}^{n\times n}$, $\mathcal{B} \in \mathbb{R}^{n\times m}$, $\mathcal{C} \in \mathbb{R}^{p\times n}$, and $x(t) = \operatorname{col}(x_1(t), x_2(t), \ldots, x_k(t)) \in \mathbb{R}^n$. Let $G_i(s) = C_i(sE_i - A_i)^{-1}B_i$ and $G_{\mathrm{cl}}(s) = \mathcal{C}(s\mathcal{E} - \mathcal{A})^{-1}\mathcal{B}$ be the transfer functions of the subsystems and the coupled (closed-loop) system. The authors prove (Corollary 2.3 in [2]) the following sufficient condition for stability of the coupled system (2).

Theorem 2.1 *Let $\Psi = \Phi_2 \operatorname{diag}(\|G_1\|_{\mathcal{H}_\infty}, \|G_2\|_{\mathcal{H}_\infty}, \ldots, \|G_k\|_{\mathcal{H}_\infty}) \in \mathbb{R}^{k\times k}$, with $\Phi_2 = [\|K_{ij}\|_2]_{i,j=1}^k$. If $\rho(\Psi) < 1$ ($\rho(\Psi)$ is the spectral radius of Ψ), then the system* (2) *is well-posed and stable.*

Monshizadeh et al. [1] find a similar condition for multi-agent systems and propose using Bounded Real Balanced Truncation (BRBT) to preserve it in the reduced order model (ROM). In this way, stability is preserved in the ROM. BRBT is a modification of the standard Balanced Truncation method, originally developed to preserve strict bounded realness, or equivalently the bound $\|G\|_{\mathcal{H}_\infty} < 1$, of the original system [3]. Algorithm 1 describes a straightforward extension of BRBT which preserves the bound $\|G\|_{\mathcal{H}_\infty} < \gamma$. The following theorem presents some properties of BRBT and an $\mathcal{H}_\infty$-error bound, and is also a straightforward extension of results in [3].

Theorem 2.2 *Let $G(s) = C(sE - A)^{-1}B$ be an LTI system of order n such that $\|G\|_{\mathcal{H}_\infty} < \gamma$. Then its ROM $\widetilde{G}(s) = \widetilde{C}(s\widetilde{E} - \widetilde{A})^{-1}\widetilde{B}$ of order $r < n$ obtained by BRBT is stable, $\|\widetilde{G}\|_{\mathcal{H}_\infty} < \gamma$, and $\|G - \widetilde{G}\|_{\mathcal{H}_\infty} \leqslant 2\gamma \sum_{i=r+1}^n \xi_i$, where ξ_i are the* bounded real singular values *(singular values of PE^TQE in Algorithm 1).*

The following Lemma is the central result which allows us to extend the idea from [1] for preserving the sufficient condition for stability by applying BRBT to subsystems of the coupled system (1).

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Algorithm 1 Bounded Real Balanced Truncation [3]

Input: LTI system $G(s) = C(sE - A)^{-1}B$ such that $\|G\|_{\mathcal{H}_\infty} < \gamma$.
Output: ROM $\widetilde{G}(s) = \widetilde{C}(s\widetilde{E} - \widetilde{A})^{-1}\widetilde{B}$.
1: Solve $APE^T + EPA^T + \frac{1}{\gamma}BB^T + \frac{1}{\gamma}EPC^TCPE^T = 0$, $A^TQE + E^TQA + \frac{1}{\gamma}C^TC + \frac{1}{\gamma}E^TQBB^TQE = 0$.
2: Find a balancing transformation T for P and E^TQE.
3: Truncate the bounded real balanced system.

Lemma 2.3 *Let $\rho(\Psi) < 1$. If $\|\widetilde{G}_i\|_{\mathcal{H}_\infty} < \frac{1}{\rho(\Psi)}\|G_i\|_{\mathcal{H}_\infty}$, for all $i \in \{1, 2, \ldots, k\}$, then $\rho(\widetilde{\Psi}) < 1$.*

Clearly, $\|G_i\|_{\mathcal{H}_\infty} < \frac{1}{\rho(\Psi)}\|G_i\|_{\mathcal{H}_\infty}$ if $\rho(\Psi) < 1$ and $G_i \neq 0$, for all $i \in \{1, 2, \ldots, k\}$. Thus, the idea is to apply BRBT to G_i with $\gamma_i = \frac{1}{\rho(\Psi)}\|G_i\|_{\mathcal{H}_\infty}$. Using Theorem 2.2, we find that

$$\|G_i - \widetilde{G}_i\|_{\mathcal{H}_\infty} \leqslant \frac{2}{\rho(\Psi)}\|G_i\|_{\mathcal{H}_\infty} \sum_{j=r_i+1}^{n_i} \xi_j^{(i)}, \tag{3}$$

where r_i is the order of $\widetilde{G}_i$ and $\xi_1^{(i)}, \xi_2^{(i)}, \ldots, \xi_{n_i}^{(i)}$ are the bounded real singular values of G_i. From results of Reis and Stykel [2], we find the upper bound for the $\mathcal{H}_\infty$-error from reducing subsystems by BRBT in the coupled system (1).

Theorem 2.4 *Denote $\delta = \frac{2}{\rho(\Psi)} \max_{i=1,2,\ldots,k} \|G_i\|_{\mathcal{H}_\infty} \sum_{j=r_i+1}^{n_i} \xi_j^{(i)}$, the maximum of bounds from* (3)*. Furthermore, let $G(s) = \operatorname{diag}(G_1(s), G_2(s), \ldots, G_k(s))$, $K = [K_{ij}]_{i,j=1}^k$, $R = \begin{bmatrix} R_1 & R_2 & \cdots & R_k \end{bmatrix}$, $H = \begin{bmatrix} H_1^T & H_2^T & \cdots & H_k^T \end{bmatrix}^T$, $g = \|K(I - GK)^{-1}\|_{\mathcal{H}_\infty}$, $g_1 = \|R(I - GK)^{-1}\|_{\mathcal{H}_\infty}$, $g_2 = \|(I - KG)^{-1}H\|_{\mathcal{H}_\infty}$, $c_1 = g_1(\|H\|_2 + g\|GH\|_{\mathcal{H}_\infty})$, and $c_2 = g_2(\|R\|_2 + g\|RG\|_{\mathcal{H}_\infty})$. If $g\delta < 1$, then*

$$\|G_{\text{cl}} - \widetilde{G}_{\text{cl}}\|_{\mathcal{H}_\infty} \leqslant \min\{c_1, c_2\}\frac{\delta}{1 - g\delta}. \tag{4}$$

Theorem 2.4 enables us to adaptively choose reduced orders $r_1, r_2, \ldots, r_k$ of the subsystems. Let $\varepsilon > 0$ be a tolerance for the error $\|G_{\text{cl}} - \widetilde{G}_{\text{cl}}\|_{\mathcal{H}_\infty}$. From (4), we see that the bound $\|G_{\text{cl}} - \widetilde{G}_{\text{cl}}\|_{\mathcal{H}_\infty} \leqslant \varepsilon$ can be achieved by enforcing $\min\{c_1, c_2\}\frac{\delta}{1-g\delta} \leqslant \varepsilon$, which is equivalent to $\delta \leqslant \frac{\varepsilon}{\min\{c_1,c_2\}+g\varepsilon}$. Finding minimal r_i such that the right hand side in (3) is less than or equal to $\frac{\varepsilon}{\min\{c_1,c_2\}+g\varepsilon}$ guarantees that $\|G_{\text{cl}} - \widetilde{G}_{\text{cl}}\|_{\mathcal{H}_\infty} \leqslant \varepsilon$. Note that $g \cdot \frac{\varepsilon}{\min\{c_1,c_2\}+g\varepsilon} < 1$, which is the assumption in Theorem 2.4.

3 Numerical example

We use the coupled string-beam example from [2], illustrated in Figure 1. After discretizing the associated partial differential equation by finite differences, the string subsystem has $n_1 = 1006$ states, $m_1 = 3$ inputs, and $p_1 = 2$ outputs, while the beam subsystem has $n_2 = 1006$ states, $m_2 = 2$ inputs, and $p_2 = 2$ outputs. Here $\rho(\Psi) = 0.9157 < 1$. Setting the reduction error tolerance for the coupled system to $\varepsilon = 10^{-3}$, we find the subsystems' error tolerance $\frac{\varepsilon}{\min\{c_1,c_2\}+g\varepsilon} = 8.5821 \cdot 10^{-5}$. Applying BRBT in Algorithm 1 and adaptive reduced order decision, we determine reduced orders $r_1 = 29$ and $r_2 = 13$. We find that $\|G_1 - \widetilde{G}_1\|_{\mathcal{H}_\infty} = 1.8858 \cdot 10^{-5}$, $\|G_2 - \widetilde{G}_2\|_{\mathcal{H}_\infty} = 1.5173 \cdot 10^{-5}$, and $\|G_{\text{cl}} - \widetilde{G}_{\text{cl}}\|_{\mathcal{H}_\infty} = 2.1345 \cdot 10^{-5} \leqslant 10^{-3}$. Figure 2 shows magnitude Bode plots of G_{cl} and $\widetilde{G}_{\text{cl}}$, demonstrating an excellent match over a wide range of frequencies.

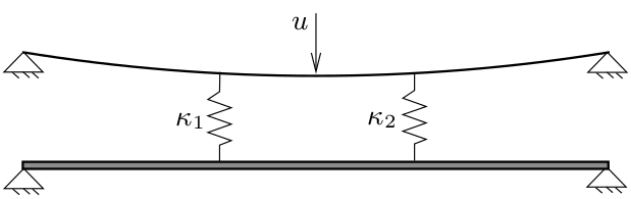

Fig. 1: Coupled string-beam example from [2].

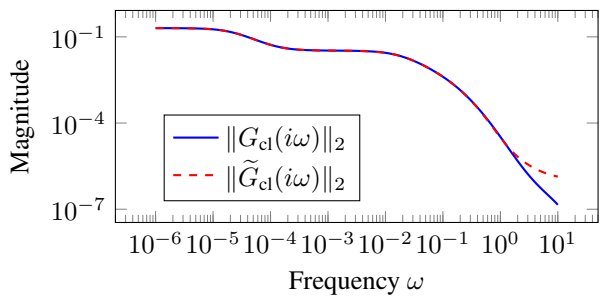

Fig. 2: Magnitude Bode plots for G_{cl} and $\widetilde{G}_{\text{cl}}$.

References

[1] N. Monshizadeh, H. L. Trentelman, and M. K. Çamlibel, Systems Control Lett. **62**(1), 1–10 (2013).
[2] T. Reis and T. Stykel, Math. Comput. Model. Dyn. Syst. **13**(5), 413–436 (2007).
[3] P. C. Opdenacker and E. A. Jonckheere, IEEE Trans. Circuits Syst. **35**(2), 184–189 (1988).

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